



MahberConnect

Research Project

Digitizing Indigenous Community Associations in Ethiopia

Made with **GAMMA**

The Problem MahberConnect Solves

Mahber associations are foundational pillars of social, cultural, and financial life in Ethiopia — yet the majority still depend on manual, paper-based processes. MahberConnect is a scalable, cloud-based SaaS platform built exclusively as a mobile application to digitize and streamline the administrative and communication workflows of these diverse associations.



Poor Communication

Manual systems make it difficult to disseminate real-time updates, leading to reduced member engagement and missed events.



Financial Opacity

Maintaining accurate financial data and ensuring transparency across contributions and expenditures is cumbersome and prone to error, eroding member trust.



Limited Scalability

As Mahber communities grow or become geographically dispersed — including diaspora — paper-based methods become unsustainable and limit reach.



Proposed Solution

A cloud-based mobile platform integrating member management, event scheduling, financial tracking, and secure internal communication.

Cont...



Excessive Administrative Burden

Leaders and volunteers spend countless hours managing membership lists, collecting dues, organizing events, and distributing information manually. This heavy workload drains valuable time and energy, often leading to burnout and limited capacity for strategic initiatives or community outreach.



Insecure & Inaccessible Records

Physical records are susceptible to loss, damage, or unauthorized access. Critical member data, financial history, and cultural documentation are vulnerable, making it difficult to recover information in case of disaster and restricting access for legitimate needs, especially for dispersed members.

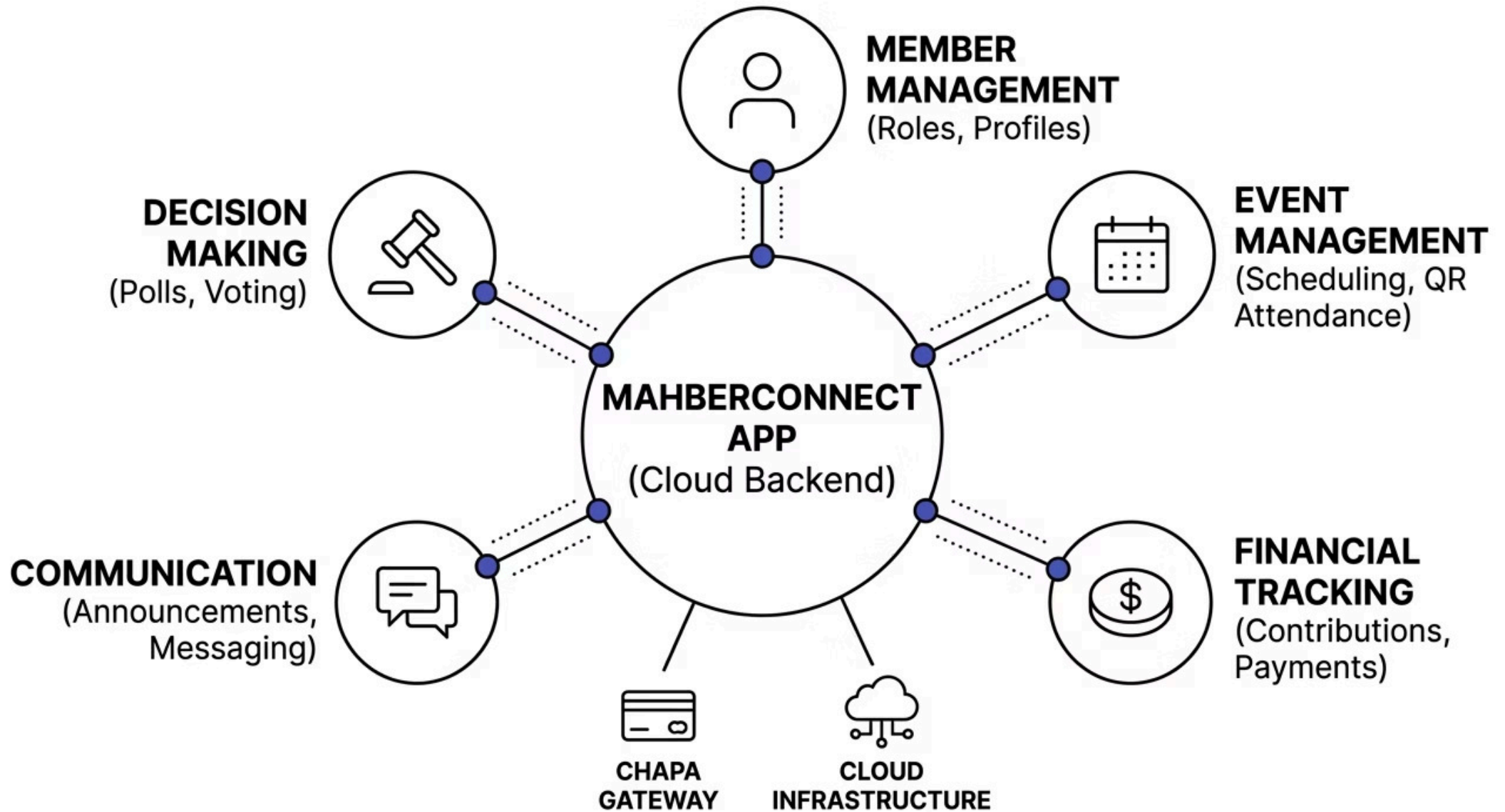


Limited Engagement & Outreach

Without modern digital platforms, Mahbers struggle to engage younger generations or reach members in the diaspora effectively. This limits potential growth, cultural exchange, and the infusion of new ideas, threatening the long-term sustainability and vibrancy of these vital community structures.

MahberConnect System Overview

MahberConnect is a unified mobile platform that brings together all aspects of Mahber management, offering a comprehensive and integrated solution to streamline operations and enhance community engagement.





OBJECTIVES

Project Goals & Specific Objectives

The goal is to build a scalable cloud platform that digitizes Mahber management in Ethiopia — improving efficiency, transparency, and member engagement while preserving cultural values.

Functional Objectives

- Digitize requirements for religious, social, and economic Mahber workflows across diverse associations.
- Centralize member profiles, roles, and permissions in one unified management system.
- Automate event scheduling, QR code check-in, and real-time member notifications.

Technical Objectives

- Develop a cloud-based architecture that supports multi-location Mahbers and nationwide or diaspora users.
- Implement automated financial tracking and reporting for contributions, expenditures, and records.
- Integrate secure local payment gateways, including **Chapa**, for convenient digital transactions.

Impact Objectives

- Increase transparency through accurate, traceable financial records and reporting.
- Improve engagement by enabling timely communication and easier participation in association activities.
- Establish a sustainable revenue model to support long-term platform operation and growth.

In summary: MahberConnect will provide a practical, scalable, and culturally aligned digital system that makes Mahber associations easier to manage, easier to trust, and easier to grow.

Functional Objectives

Digitize Mahber workflows, centralize member management, and automate scheduling, check-ins, and notifications.

Technical Objectives

Build a cloud-based system with financial tracking, reporting, and secure local payment integration.

Impact Objectives

Improve transparency, boost member engagement, and create a sustainable model for long-term growth.

Agile Scrum Development & Tech Stack

The project adopted the Agile Scrum methodology, utilizing a modern, performant technology stack to build a cross-platform mobile experience. Development is structured in focused phases ensuring iterative delivery and continuous stakeholder alignment.

1 Phase 1: Requirements & Discovery

Gather, analyze, and specify functional and non-functional requirements from stakeholder interviews with religious, social, and economic Mahber associations.

2 Phase 2: Architecture & Design

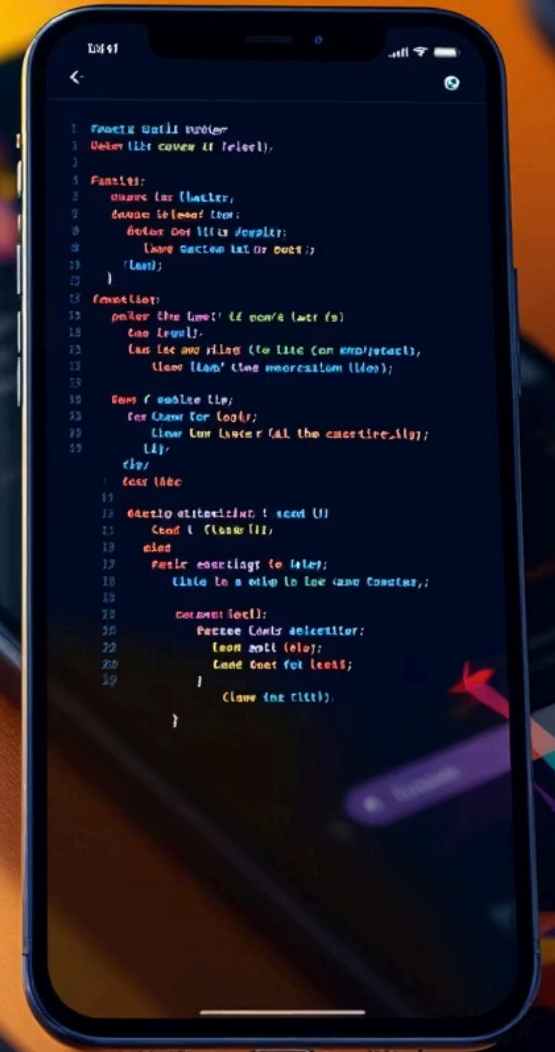
Design Go microservices backend with PostgreSQL data store. Plan Flutter frontend for cross-platform mobile delivery. Architect cloud-based, scalable SaaS infrastructure.

3 Phase 3: Core Module Development

Build member management, event scheduling with QR attendance, financial tracking, and group communication modules. Integrate Chapa payment gateway.

4 Phase 4: Testing & Deployment

Conduct rigorous testing, validate functional requirements, deploy to cloud infrastructure, and document lessons learned for future scalability improvements.



MahberConnect Feature Architecture

MahberConnect's modular design ensures every dimension of Mahber community life is addressed in a single, unified platform — from membership and finance to events and governance. Each module directly closes a gap identified in the literature review.



Unlike prior systems that digitized only financial functions, MahberConnect unifies membership, finance, events, communication, and governance into one culturally adaptive, cloud-based mobile platform for Ethiopia's indigenous associations.



LITERATURE REVIEW

Gaps in Existing Solutions

A review of existing digital platforms — including Digital Equb, ChamaApp, Wild Apricot, and SaccoTech — reveals that while digitization of cooperative finance has advanced, critical gaps persist that MahberConnect is specifically designed to fill.

1

Narrow Financial Focus

Most systems only automate savings and contributions, overlooking event coordination, attendance, role assignment, and communication — all central to Mahber life.

2

No Cultural Adaptability

Imported platforms lack the flexibility for local language, rotational hosting, religious ceremonies, and hierarchical social roles unique to Ethiopian associations.

3

Single-Org Limitation

Existing tools support only one organization at a time, yet many Ethiopians belong to multiple Mahbers simultaneously — a complexity no current platform handles effectively.



FUNCTIONAL REQUIREMENTS

Key Functional Requirements Summary

The following core requirement areas define what MahberConnect must do to effectively serve diverse Mahber associations. All requirements are classified as Mandatory for initial launch.



Member & Multi-Mahber Management

Role-based access (Admin, Treasurer, Host, Advisor, Member), membership approval workflows, member banning controls, and seamless switching between multiple Mahbers without logging out (FR-MEM-001 to FR-MEM-006).



Event Management & Communication

Event creation and scheduling, invitation dispatch via in-app, SMS, or email, QR code attendance tracking, group messaging, announcement management, and multi-channel notifications (FR-EVT-001 to FR-EVT-006, FR-COM-001 to FR-COM-004).



Finance, Transparency & Decision-Making

Contribution and payout management, digital receipts, Chapa gateway integration, financial data export, activity audit trails, poll and voting functionality, and result storage for accountability (FR-FIN-001 to FR-FIN-005, FR-DEC-001 to FR-DEC-003, FR-TRN-001 to FR-TRN-002).

MahberConnect is scoped as a mobile-only application with Amharic and English support at launch — web and desktop versions, advanced AI features, and extended language support are planned for future iterations.